Inductive real-interval covers agree with pointwise covers under excluded middle

The Univalent Foundations Program

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Editorial note. Translate between proof-generated covers and pointwise covers: induction on cover constructors in one direction and a Lebesgue-number subdivision with repeated transitivity in the other. Existing unpublished S046 H2 judgment is reused.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained. Classical finite-cover Lebesgue-number lemma; this is the explicitly cited standard input, not an omitted target construction.

Source reference. Original reference 11.12: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1358–1358. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

to be a map

$$\mathcal{F} : I \to \mathbb{Q} \times \mathbb{Q}.$$

To be quite precise, a family is a dependent pair $(I, \mathcal{F})$, not just $\mathcal{F}$. A **finite family of basic intervals** is one indexed by $\{\, m : \mathbb{N} \mid m < n \,\}$ for some $n : \mathbb{N}$. We usually present it by a finite list $[(q_0, r_0), \ldots, (q_{n-1}, r_{n-1})]$. Finally, a **finite subfamily** of $(I, \mathcal{F})$ is given by a list of indices $[i_1, \ldots, i_n]$ which then determine the finite family $[\mathcal{F}(i_1), \ldots, \mathcal{F}(i_n)]$.

As long as we are aware of the distinction between a pair (q, r) and the corresponding interval $\{\, x : \mathbb{R} \mid q < x < r \,\}$, we may safely use the same notation (q, r) for both. Intersections and inclusions of intervals are expressible in terms of their endpoints:

$$\begin{aligned} (q, r) \cap (s, t) &:\equiv (\max(q, s), \min(r, t)), \\ (q, r) \subseteq (s, t) &:\equiv (q < r \Rightarrow s \leq q < r \leq t). \end{aligned}$$

We say that $(I, \lambda i.\,(q_i, r_i))$ **(pointwise) covers** $[a, b]$ when

$$\forall(x : [a, b]).\,\exists(i : I).\,q_i < x < r_i. \tag{11.0.1}$$

The **Heine–Borel compactness for** $[0, 1]$ states that every covering family of $[0, 1]$ merely has a finite subfamily which still covers $[0, 1]$.

Theorem 11.0.2. *If excluded middle holds then* $[0, 1]$ *is Heine–Borel compact.*

Proof. Assume for the purpose of reaching a contradiction that a family $(I, \lambda i.\,(a_i, b_i))$ covers $[0, 1]$ but no finite subfamily does. We construct a sequence of closed intervals $[q_n, r_n]$ which are nested, their sizes shrink to 0, and none of them is covered by a finite subfamily of $(I, \lambda i.\,(a_i, b_i))$.

We set $[q_0, r_0] :\equiv [0, 1]$. Assuming $[q_n, r_n]$ has been constructed, let $s :\equiv (2q_n + r_n)/3$ and $t :\equiv (q_n + 2r_n)/3$. Both $[q_n, t]$ and $[s, r_n]$ are covered by $(I, \lambda i.\,(a_i, b_i))$, but they cannot both have a finite subcover, or else so would $[q_n, r_n]$. Either $[q_n, t]$ has a finite subcover or it does not. If it does we set $[q_{n+1}, r_{n+1}] :\equiv [s, r_n]$, otherwise we set $[q_{n+1}, r_{n+1}] :\equiv [q_n, t]$.

The sequences $q_0, q_1, \ldots$ and $r_0, r_1, \ldots$ are both Cauchy and they converge to a point $x : [0, 1]$ which is contained in every $[q_n, r_n]$. There merely exists $i : I$ such that $a_i < x < b_i$. Because the sizes of the intervals $[q_n, r_n]$ shrink to zero, there is $n : \mathbb{N}$ such that $a_i < q_n \leq x \leq r_n < b_i$, but this means that $[q_n, r_n]$ is covered by a single interval (a_i, b_i), while at the same time it has no finite subcover. A contradiction. □

Definition 11.0.3. The **inductive cover** $\lhd$ is a mere relation

$$\lhd : (\mathbb{Q} \times \mathbb{Q}) \to \Big(\sum_{I:\mathcal{U}} (I \to \mathbb{Q} \times \mathbb{Q})\Big) \to \mathsf{Prop}$$

defined inductively by the following rules, where q, r, s, t are rational numbers and $(I, \mathcal{F})$, $(J, \mathcal{G})$ are families of basic intervals:

(i) *reflexivity:* $\mathcal{F}(i) \lhd (I, \mathcal{F})$ for all $i : I$,
(ii) *transitivity:* if $(q, r) \lhd (J, \mathcal{G})$ and $\forall(j : J).\,\mathcal{G}(j) \lhd (I, \mathcal{F})$ then $(q, r) \lhd (I, \mathcal{F})$,
(iii) *monotonicity:* if $(q, r) \subseteq (s, t)$ and $(s, t) \lhd (I, \mathcal{F})$ then $(q, r) \lhd (I, \mathcal{F})$,
(iv) *localization:* if $(q, r) \lhd (I, \mathcal{F})$ then $(q, r) \cap (s, t) \lhd (I, \lambda i.\,(\mathcal{F}(i) \cap (s, t)))$.
(v) if $q < s < t < r$ then $(q, r) \lhd [(q, t), (s, r)]$,
(vi) $(q, r) \lhd (\{\, (s, t) : \mathbb{Q} \times \mathbb{Q} \mid q < s < t < r \,\}, \lambda u.\,u)$.

The definition should be read as a higher-inductive type in which the listed rules are point constructors, and the type is (-1)-truncated. The first four clauses are of a general nature and should be intuitively clear. The last two clauses are specific to the real line: one says that an interval may be covered by two intervals if they overlap, while the other one says that an interval may be covered from within. Incidentally, if $r \leq q$ then (q, r) is covered by the empty family by the last clause.

Inductive covers enjoy the Heine–Borel property, the proof of which requires a lemma.

Lemma 11.0.4. *Suppose* $q < s < t < r$ *and* $(q, r) \lhd (I, \mathcal{F})$. *Then there merely exists a finite subfamily of* $(I, \mathcal{F})$ *which inductively covers* (s, t).

Proof. We prove the statement by induction on $(q, r) \lhd (I, \mathcal{F})$. There are six cases:

(i) Reflexivity: if $(q, r) = \mathcal{F}(i)$ then by monotonicity (s, t) is covered by the finite subfamily $[\mathcal{F}(i)]$.

(ii) Transitivity: suppose $(q, r) \lhd (J, \mathcal{G})$ and $\forall(j : J).\, \mathcal{G}(j) \lhd (I, \mathcal{F})$. By the inductive hypothesis there merely exists $[\mathcal{G}(j_1), \ldots, \mathcal{G}(j_n)]$ which covers (s, t). Again by the inductive hypothesis, each of $\mathcal{G}(j_k)$ is covered by a finite subfamily of $(I, \mathcal{F})$, and we can collect these into a finite subfamily which covers (s, t).

(iii) Monotonicity: if $(q, r) \subseteq (u, v)$ and $(u, v) \lhd (I, \mathcal{F})$ then we may apply the inductive hypothesis to $(u, v) \lhd (I, \mathcal{F})$ because $u < s < t < v$.

(iv) Localization: suppose $(q', r') \lhd (I, \mathcal{F})$ and $(q, r) = (q', r') \cap (a, b)$. Because $q' < s < t < r'$, by the inductive hypothesis there is a finite subcover $[\mathcal{F}(i_1), \ldots, \mathcal{F}(i_n)]$ of (s, t). We also know that $a < s < t < b$, therefore $(s, t) = (s, t) \cap (a, b)$ is covered by $[\mathcal{F}(i_1) \cap (a, b), \ldots, \mathcal{F}(i_n) \cap (a, b)]$, which is a finite subfamily of $(I, \lambda i.\, (\mathcal{F}(i) \cap (a, b)))$.

(v) If $(q, r) \lhd [(q, v), (u, r)]$ for some $q < u < v < r$ then by monotonicity $(s, t) \lhd [(q, v), (u, r)]$.

(vi) Finally, $(s, t) \lhd (\{\, (u, v) : \mathbb{Q} \times \mathbb{Q} \mid q < u < v < r \,\}, \lambda z.\, z)$ by reflexivity. □

Say that $(I, \mathcal{F})$ **inductively covers** $[a, b]$ when there merely exists $\epsilon : \mathbb{Q}_+$ such that $(a - \epsilon, b + \epsilon) \lhd (I, \mathcal{F})$.

Corollary 11.0.5. *A closed interval is Heine–Borel compact for inductive covers.*

Proof. Suppose $[a, b]$ is inductively covered by $(I, \mathcal{F})$, so there merely is $\epsilon : \mathbb{Q}_+$ such that $(a - \epsilon, b + \epsilon) \lhd (I, \mathcal{F})$. By Lemma 11.0.4 there is a finite subcover of $(a - \epsilon/2, b + \epsilon/2)$, which is therefore a finite subcover of $[a, b]$. □

Experience from formal topology shows that the rules for inductive covers are sufficient for a constructive development of pointfree topology. But we can also provide our own evidence that they are a reasonable notion.

Theorem 11.0.6.

(i) An inductive cover is also a pointwise cover.

(ii) Assuming excluded middle, a pointwise cover is also an inductive cover.

Proof.

(i) Consider a family of basic intervals $(I, \mathcal{F})$, where we write $(q_i, r_i) :\equiv \mathcal{F}(i)$, an interval (a, b) inductively covered by $(I, \mathcal{F})$, and x such that $a < x < b$. We prove by induction on $(a, b) \lhd (I, \mathcal{F})$ that there merely exists $i : I$ such that $q_i < x < r_i$. Most cases are pretty obvious, so we show just two. If $(a, b) \lhd (I, \mathcal{F})$ by reflexivity, then there merely is some $i : I$ such that $(a, b) = (q_i, r_i)$ and so $q_i < x < r_i$. If $(a, b) \lhd (I, \mathcal{F})$ by transitivity via $(J, \lambda j.\, (s_j, t_j))$ then by the inductive hypothesis there merely is $j : J$ such that $s_j < x < t_j$, and then since $(s_j, t_j) \lhd (I, \mathcal{F})$ again by the inductive hypothesis there merely exists $i : I$ such that $q_i < x < r_i$. Other cases are just as exciting.

(ii) Suppose $(I, \lambda i.\,(q_i, r_i))$ pointwise covers (a, b). By item (vi) of Definition 11.0.3 it suffices to show that $(I, \lambda i.\,(q_i, r_i))$ inductively covers (c, d) whenever $a < c < d < b$, so consider such c and d. By Theorem 11.0.2 there is a finite subfamily $[i_1, \ldots, i_n]$ which already pointwise covers $[c, d]$, and hence (c, d). Let $\epsilon : \mathbb{Q}_+$ be a Lebesgue number for $(q_{i_1}, r_{i_1}), \ldots, (q_{i_n}, r_{i_n})$ as in original reference 11.12. There is a positive $k : \mathbb{N}$ such that $2(d - c)/k < \min(1, \epsilon)$. For $0 \leq i \leq k$ let

$$c_k :\equiv ((k - i)c + id)/k.$$

The intervals (c_0, c_2), (c_1, c_3), ..., (c_{k-2}, c_k) inductively cover (c, d) by repeated use of transitivity and item (v) in Definition 11.0.3. Because their widths are below ϵ each of them is contained in some (q_i, r_i), and we may use transitivity and monotonicity to conclude that $(I, \lambda i.\,(q_i, r_i))$ inductively cover (c, d). □